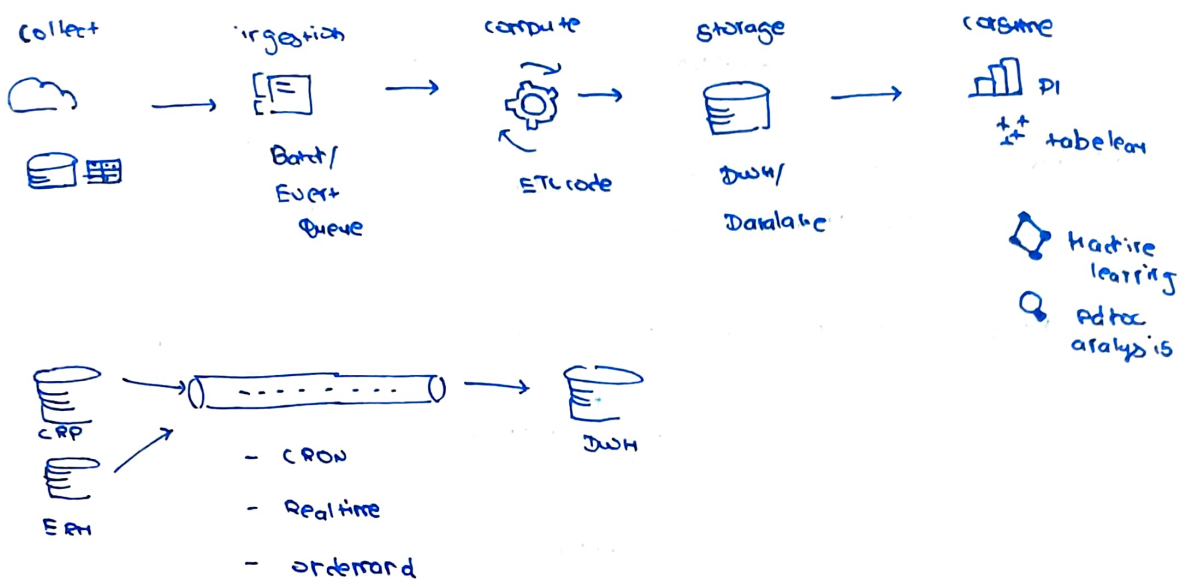
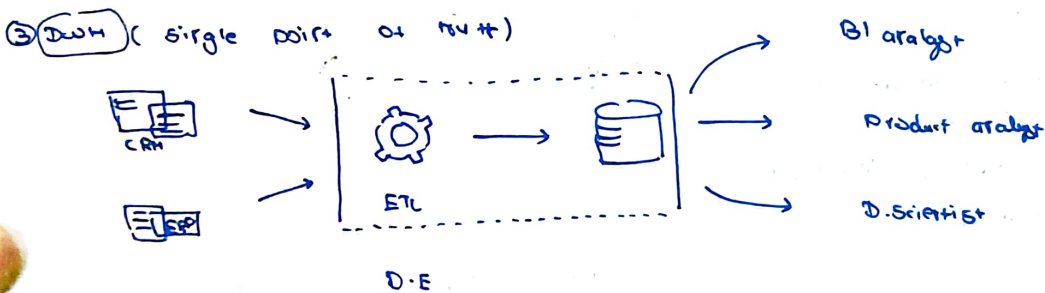
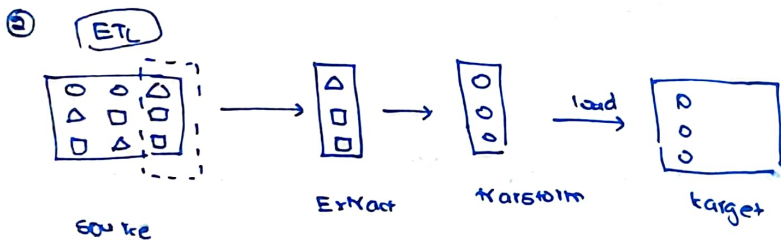


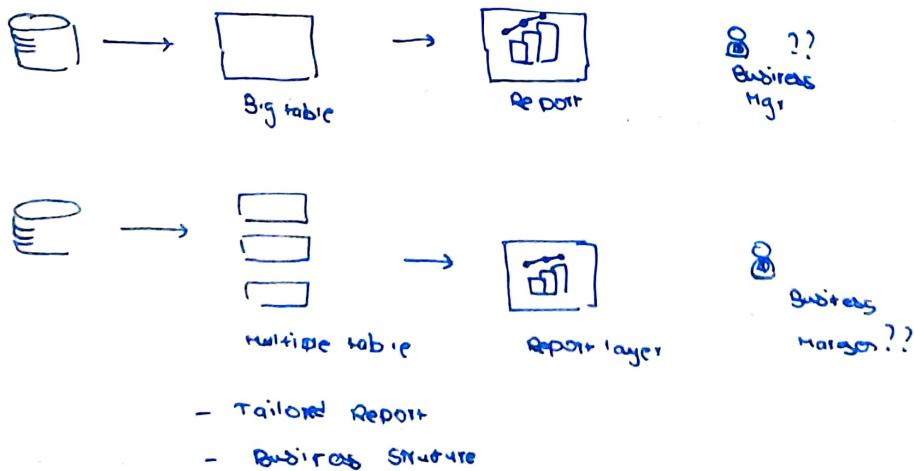
① Data pipeline :: Flow of data from one place to another



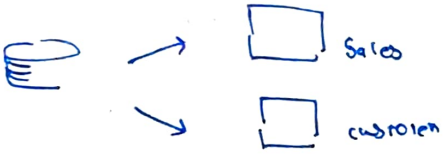
- 150% zoom



④ Data Modeling :: Business data & how the pieces are related



DBA dim



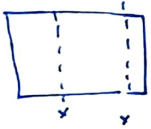
8 Data analyst ??

8 Financial analyst

- How to join
- what is what?

Problem :: Hard to maintain || Redundancy performance issue

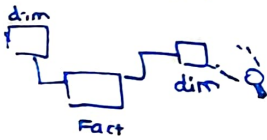
Physical layer || Model layer || Report layer



- ↳ Relationship
- ↳ cardinality (Mathematical context)
- ↳ Filter

Zoom [-+] 150%

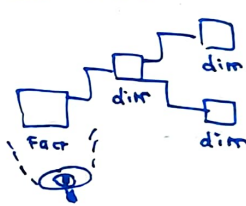
Star schema ☆



Fact table sits at the middle with multiple dimensions surrounding

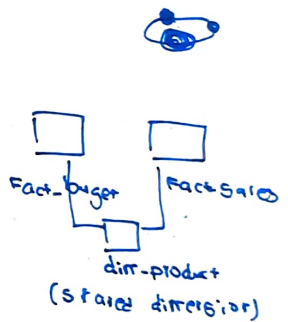
- ↳ Easy to read
- ↳ Big Dimension

|| Snowflake schema ☆

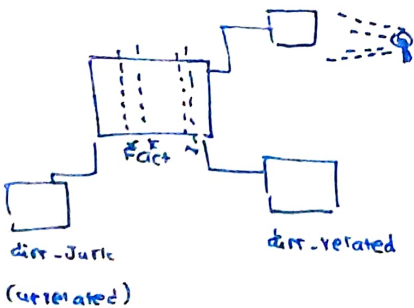


- ↳ complex (Require multiple Joins)
- ↳ storage

|| Galaxy schema



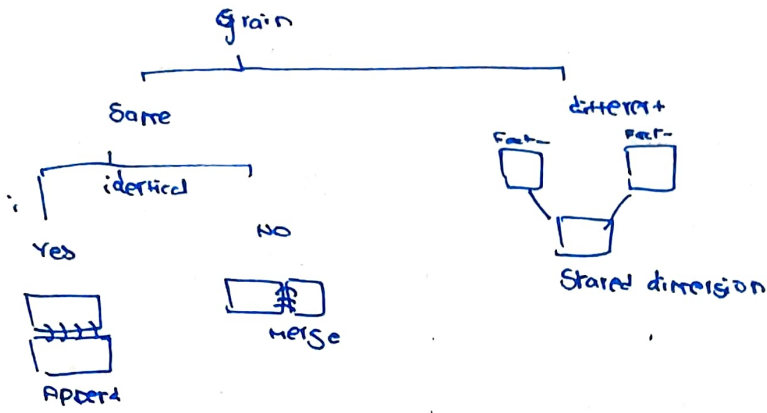
with business it is impossible to have only one Fact table



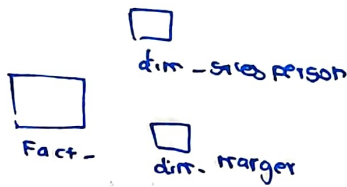
Grain || level of info in each row



dim-employees (role playing dimension)



why?



Don't go around create dimension table for every contexts.

Fact-table

"

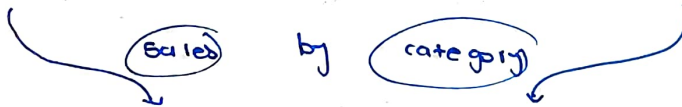
Dimension-table

- Quantitative info
 - Events
 - Transactions

- Descriptive info correlated
 to the action fact it gives
 more context

core: Id ; date ; measure

core ; ; attributes



⑤ Data governance :: creating rule to protect our data

Data life cycle ::



Full acquisition



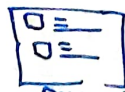
Disposal

key components ::



Policy

What are the rules?



Procedure

How to implement the rules?

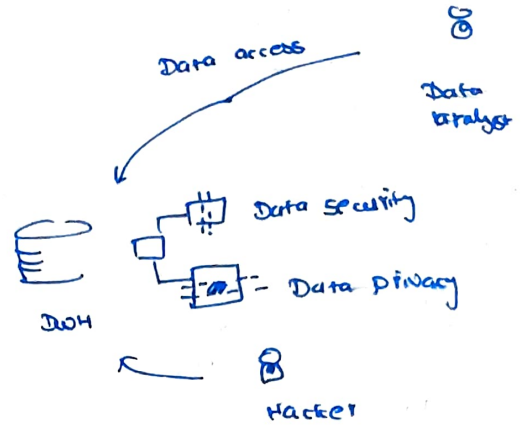
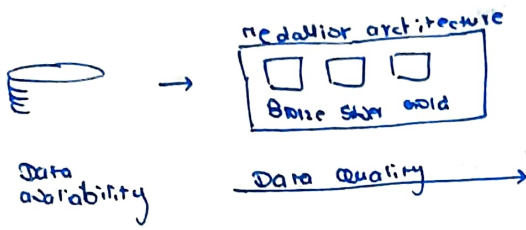


Roles

Who is responsible
 for what?

Some common policy ::

- Data access
- Data Security
- Data Privacy
- Data Quality
- Data Retention



⑥ Data Quality :: can we trust our data

- ↳ completeness
- ↳ uniqueness
- ↳ validation
- ↳ correctness

